

COMMONWEALTH OF AUSTRALIA

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FIT3155: Advanced Algorithms and Data Structures

Week 10: Linear Programming

Faculty of Information Technology, Monash University

Linear Programming introduction

- Arose out of efforts dating back to World War-II when Mathematicians were asked to solve logistics problems for military operations.
- Linear Programming is a combinatorial optimization method to operate specific systems **optimally** under resource **constraints**.
- It was introduced by **Dantzig** in 1947 in a paper entitled **“Programming in Linear Structure”**
- A method of solving such problems (**simplex method**) was proposed in 1949.

Dantzig



“George Dantzig will go down in history as one of the founders and chief contributors to the field of mathematical programming, and as the creator of the simplex algorithm for linear programming, perhaps the most important algorithm developed in the 20th century.” – Richard Karp.

Consider this problem

- A small manufacturer makes two products A and B
- Each of these products require resources R_1 and R_2 to make
- Each unit of A requires **1 unit** of R_1 and **3 units** of R_2
- Each unit of B requires **1 unit** of R_1 and **2 units** of R_2
- But the manufacturer has only **5 units** of R_1 and **12 units** of R_2
- On every unit of A sold the manufacturer makes \$6 **profit**
- On every unit of B sold the manufacturer makes \$5 **profit**

Problem statement

How many units of the product A and B should the manufacturer produce to **maximize** profit?

Formalizing the problem

Problem statements

- A small manufacturer makes two products A and B
- Each of these products require resources R_1 and R_2 to make
- Each unit of A requires 1 unit of R_1 ...
- ...and 3 units of R_2
- Each unit of B requires 1 unit of R_1 ...
- ...and 2 units of R_2
- But the manufacturer has only 5 units of R_1 ...
- ...and 12 units of R_2
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- On every unit of B sold the manufacturer makes \$5 profit

Formalism

Formalizing the problem

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Formalism

x = #units of product A produced

y = #units of product B produced

Formalizing the problem

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Formalism

x = #units of product A produced

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$$z = 6x + 5y$$

Formalizing the problem

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- On every unit of B sold the manufacturer makes \$5 profit

Formalism

x = #units of product A produced

y = #units of product B produced

$$z = 6x + 5y$$

$$x + y \leq 5$$

Formalizing the problem

Problem statements

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Formalism

x = #units of product A produced

y = #units of product B produced

$$z = 6x + 5y$$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

Formalizing the problem

Problem statements

- A small manufacturer makes two products A and B
- Each of these products require resources R_1 and R_2 to make
- Each unit of A requires **1 unit** of R_1 ...
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Formalism

x = #units of product A produced

y = #units of product B produced

$$z = 6x + 5y$$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

$$x, y \geq 0$$

Formalizing the problem

Problem statements

- A small manufacturer makes two products A and B
- Each of these products require resources R_1 and R_2 to make
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Formalism

x = #units of product A produced

y = #units of product B produced

maximize (profit) $z = 6x + 5y$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

$$x, y \geq 0$$

Now let's introduce some technical jargon

Decision Variables

The variable x and y are called **decision variables**

Objective function

The profit function ($z = 6x + 5y$) that is being optimized is called the **objective function**.

Constraints

The restrictions

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

$$x, y \geq 0$$

are called (linear) **constraints**. These restrict the values the decision variables can take.

We now have a linear programming formulation

x = number of units of product A produced

y = number of units of product B produced

$$\text{maximize (profit)} \quad z = 6x + 5y$$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

$$x, y \geq 0$$

All standard Linear Programming (LP) problems have...

A **linear** objective function

A set of **linear** constraints

A **non-negativity constraint** on all decision variables

Linear Programming problem in its standard form

Maximize

$$z = c_1 * x_1 + c_2 * x_2 + \cdots + c_n * x_n$$

Subject to the constraints:

$$a_{11} * x_1 + a_{12} * x_2 + \cdots + a_{1n} * x_n \leq b_1$$

$$a_{21} * x_1 + a_{22} * x_2 + \cdots + a_{2n} * x_n \leq b_2$$

$$\vdots$$

$$a_{m1} * x_1 + a_{m2} * x_2 + \cdots + a_{mn} * x_n \leq b_m$$

and

$$x_1, x_2, \cdots x_n \geq 0$$

Gaining insights from a graphical solution – although it is **limited** in practice!

Let's **graph** the example linear programming problem to gain some insights about solutions of **general** linear programs.^a

^aA graphical solution is applicable in practice to solutions involving 2 (or even 3) decision variables.

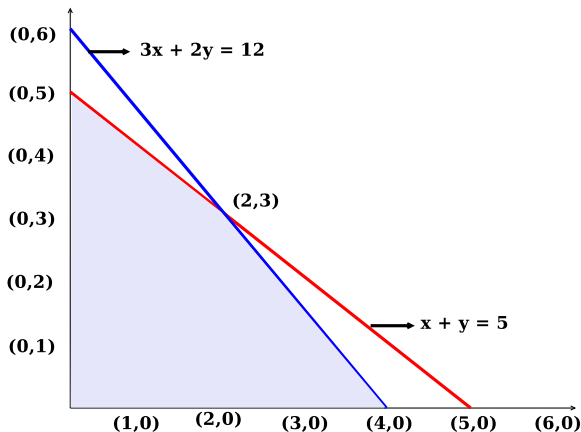
Graphical view of the example Linear Program

maximize profit: $z = 6x + 5y$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

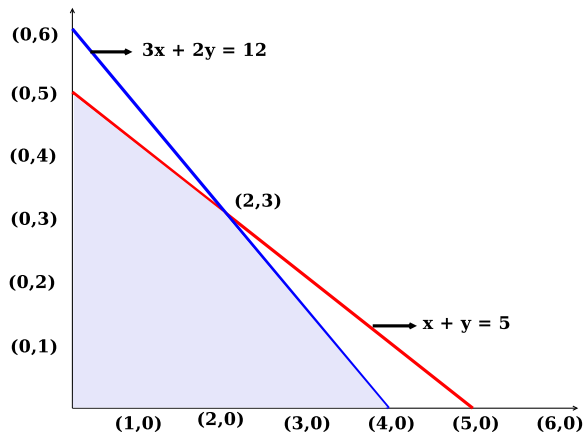
$$x, y \geq 0$$



Graphical view – Insights

Feasible region

...is the region where the given linear program has **feasible solutions** **satisfying** all its constraints

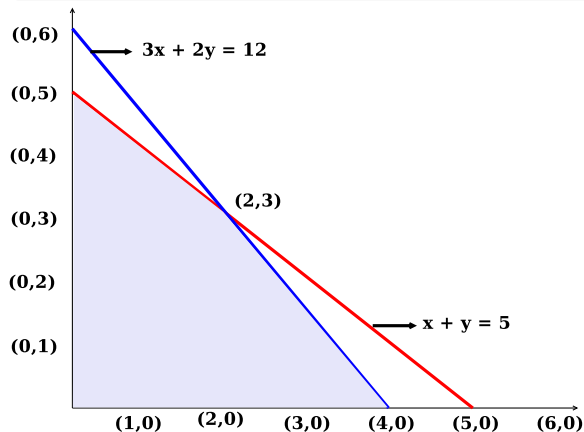


Graphical view – Insights

Insight 1

Feasible region is a **convex polyhedron**.^a

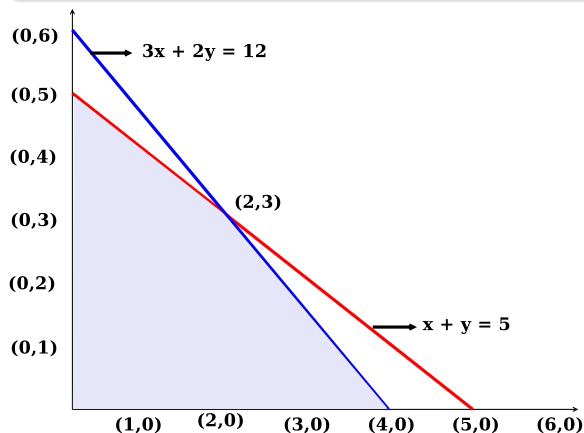
^a Given **any two points inside** or **on** the polyhedron, the line segment connecting those two points always lies entirely within the polyhedron.



Graphical view – Insights

Insight 2

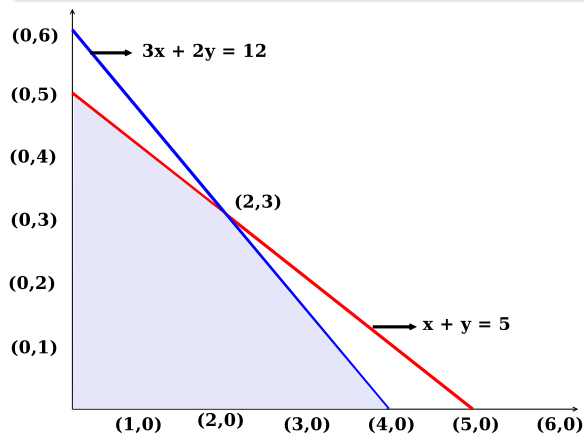
For **any point** in the **feasible region**, there is always a point **on the boundary/surface** of the feasible region that will **dominate** it (i.e., will give a better evaluation of the **objective function**)



Graphical view – Insights

Insight 3

Further, **among all points on the boundary/surface** of the feasible regions, there is **at least one corner point** of the feasible region that will always **dominate**, given the objective function.



Now let's explore this graphical view with some added algebra

- An important step here is to convert the **inequality** constraints into **equality** constraints.
- This is achieved by adding **slack variables** to convert an inequality to an equality.

Algebra (of the graphical view) – changing inequality constraints to equality constraints

By introducing **slack variables**, we get the following:

First constraint: $x + y \leq 5$ becomes

$$x + y + s = 5$$

where $s \geq 0$

Similarly the second constraint $3x + 2y \leq 12$ becomes

$$3x + 2y + t = 12$$

where $t \geq 0$

Objective function can be written as:

$$\text{Maximize } z = 6x + 5y + 0s + 0t$$

Algebra (of the graphical view) – changing inequality constraints to equality constraints

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where $t \geq 0$

Objective function can be written as:

$$\text{Maximize } z = 6x + 5y + 0s + 0t$$

Algebra (of the graphical view) – Problem formulation

Therefore, solving this problem:

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

and $x, y \geq 0$

...is same as solving this problem:

Maximize

$$z = 6x + 5y + 0s + 0t$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

Can a solution to be found in this set up?

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

This set up has 4 variables but 2 equations.

Can a solution to be found in this set up?

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

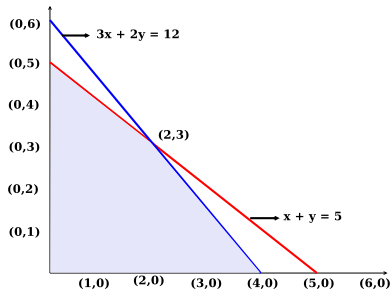
and $x, y, s, t \geq 0$

This set up has 4 variables but 2 equations.

A solution (inspired by Insight 3 on Slide 11) can be found for any 2 variables, if the remaining 2 variables are 'fixed' to 0!

Algebraic (graphical) solution – explore **all** possible ways of fixing variables

Fix x, y to 0	and solve for s, t
Fix x, s to 0	and solve for y, t
Fix x, t to 0	and solve for y, s
Fix y, s to 0	and solve for x, t
Fix y, t to 0	and solve for x, s
Fix s, t to 0	and solve for x, y



Algebraic (graphical) solution – exhaustive search

Maximize

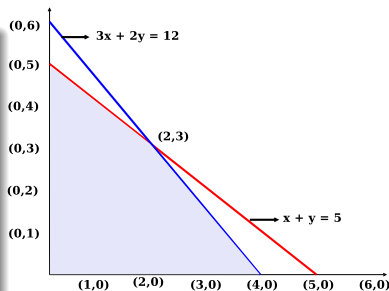
$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$



By fixing the variables in the left hand column to zeros, we get:

Non-Basic Variables	Basic Variables
Fixing $x = 0, y = 0$	We get $s = 5, t = 12$
Fixing $x = 0, s = 0$	We get $y = 5, t = 2$
Fixing $x = 0, t = 0$	We get $y = 6, s = -1$
Fixing $y = 0, s = 0$	We get $x = 5, t = -3$
Fixing $y = 0, t = 0$	We get $x = 4, s = 1$
Fixing $s = 0, t = 0$	We get $x = 2, y = 3$

Counter-intuitive terminology alert!!!

In the context of linear programming, we get to hear the following terms that might appear counter-intuitive:

Basic variables

- The **free** (i.e., **not** fixed) variables are called **basic variables**.
 - ▶ Basic variables correspond specifically to column (unit) vectors that form a **basis** (vector space) of the iterative linear programming solution – **you can ignore this part, but I will explain this discussing Tableau Simplex later.**

Non-basic variables

- **Non-basic** variables are the variable that are **fixed** to **zero**
 - ▶ They are so called, because they are **OUTSIDE** the **basis**.

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	$z = 25$
Fix $x = 0, t = 0$	$y = 6, s = -1$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	$z = 25$
Fix $x = 0, t = 0$	$y = 6, s = -1$	Infeasible
Fix $y = 0, s = 0$	$x = 5, t = -3$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	$z = 25$
Fix $x = 0, t = 0$	$y = 6, s = -1$	Infeasible
Fix $y = 0, s = 0$	$x = 5, t = -3$	Infeasible
Fix $y = 0, t = 0$	$x = 4, s = 1$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	$z = 25$
Fix $x = 0, t = 0$	$y = 6, s = -1$	Infeasible
Fix $y = 0, s = 0$	$x = 5, t = -3$	Infeasible
Fix $y = 0, t = 0$	$x = 4, s = 1$	$z = 24$
Fix $s = 0, t = 0$	$x = 2, y = 3$	

Algebraic (graphical) Solution – exhaustive search

Objective

$$z = 6x + 5y$$

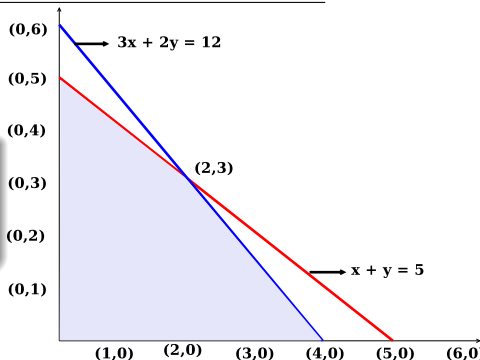
Now let's evaluate the objective function corresponding to all possible ways of choosing/fixing variables to become non-basic variables:

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	Evaluated value of objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
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Fix $x = 0, t = 0$	$y = 6, s = -1$	Infeasible
Fix $y = 0, s = 0$	$x = 5, t = -3$	Infeasible
Fix $y = 0, t = 0$	$x = 4, s = 1$	$z = 24$
Fix $s = 0, t = 0$	$x = 2, y = 3$	$z = 27$

Algebraic (graphical) solution – exhaustive search

Non-Basic Variables (fixed to 0 variables)	Basic Variables (free variables)	value of objective objective function
Fix $x = 0, y = 0$	$s = 5, t = 12$	$z = 0$
Fix $x = 0, s = 0$	$y = 5, t = 2$	$z = 25$
Fix $x = 0, t = 0$	$y = 6, s = -1$	Infeasible
Fix $y = 0, s = 0$	$x = 5, t = -3$	Infeasible
Fix $y = 0, t = 0$	$x = 4, s = 1$	$z = 24$
Fix $s = 0, t = 0$	$x = 2, y = 3$	$z = 27$

Keep in mind that the values x and y above correspond to the corner points (including those outside the feasible region)



Now generalizing this exhaustive search

- Let some general linear programming formulation contain

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.
- There are N **decision variables** and m **linear equations** in this system.

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.
- There are N **decision variables** and m **linear equations** in this system.
- Assign N out of $N + m$ (**decision+slack**) variables to 0, and solve for the remaining m variables.

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.
- There are N **decision variables** and m **linear equations** in this system.
- Assign N out of $N + m$ (**decision+slack**) variables to 0, and solve for the remaining m variables.
- There are $N+mC_N$ such possibilities, **sigh!**

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.
- There are N **decision variables** and m **linear equations** in this system.
- Assign N out of $N + m$ (**decision+slack**) variables to 0, and solve for the remaining m variables.
- There are $N+mC_N$ such possibilities, **sigh!**
- Substitute the values of that N **decision variables** take into the objective function.

Now generalizing this exhaustive search

- Let some general linear programming formulation contain
 - ▶ N **decision variables**
 - ▶ m **linear constraints**
- For m linear constraints, there will be m **slack variables**.
- There are N **decision variables** and m **linear equations** in this system.
- Assign N out of $N + m$ (**decision+slack**) variables to 0, and solve for the remaining m variables.
- There are $N+mC_N$ such possibilities, **sigh!**
- Substitute the values of that N **decision variables** take into the objective function.
- Once all possibilities are explored, choose the **optimum** over all the solutions.

Disadvantages

- Grows combinatorially for large N
- Evaluates **infeasible solutions**.
- Does not tell you if the optima has been reached. The search is exhaustive (brute-force).
- Does not give progressively better solutions (in terms of the objective function) – the search is fairly random.

Simplex method to solving linear programming problems

Simplex Method is a way to solve this problem. Advantages of simplex are:

- Polynomial time (in practice, but not in worst case)
- Does NOT evaluate infeasible solutions.
- Explores progressively better solutions.
- Tells you when the optima has been reached and stops.

Simplex Method – explored in two forms

We will now see two forms of the Simplex method using the example we have been considering:

- Algebraic form of Simplex: **this will give you the insights of how the algorithm works.**
- Tabular (or tableau) form of Simplex: **this will allow you to convert it into code.**

Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

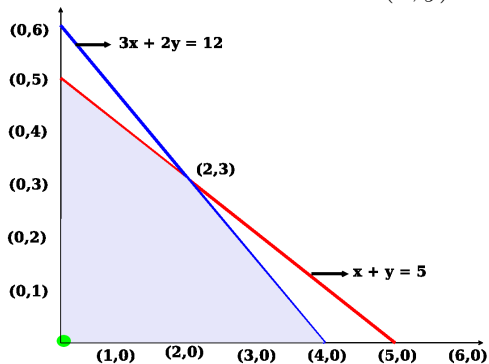
Start

Start by choosing x and y as **non-basic variables** (i.e., variables that are **set to zero**.)

This gives the solutions to s and t as $s = 5$ and $t = 12$ respectively.

Simplex method – Algebraic form

Visualize what this solution of $(x, y) = (0, 0)$ implies:



Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

First iteration

Rewrite the constraints in terms of s and t (now basic variables) each with coefficient of +1

Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

First iteration

Rewrite the constraints in terms of s and t (now basic variables) each with coefficient of +1

$$s = 5 - x - y$$

Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

First iteration

Rewrite the constraints in terms of s and t (now basic variables) each with coefficient of +1

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

Simplex method – Algebraic form

Maximize

$$z = 6x + 5y$$

Subject to the constraints:

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

and $x, y, s, t \geq 0$

First iteration

Rewrite the constraints in terms of s and t (now basic variables) each with coefficient of +1

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

Objective is simply: $z = 6x + 5y$

Simplex method – Algebraic form

Problem as we transformed it in the previous slide

Maximize

$$z = 6x + 5y$$

Subject to the constraints (notice we have rewritten the constraints as in the previous slide):

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Recall, currently $x = 0$ and $y = 0$. Therefore the objective function, substituting these values, is $z = 0$.

*To increase the objective function, if the coefficient of the variable in the objective function is positive, you increase the variable. Else, if its coefficient is negative you decrease the variable.

Simplex method – Algebraic form

Problem as we transformed it in the previous slide

Maximize

$$z = 6x + 5y$$

Subject to the constraints (notice we have rewritten the constraints as in the previous slide):

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Recall, currently $x = 0$ and $y = 0$. Therefore the objective function, substituting these values, is $z = 0$. Can we improve z ?

*To increase the objective function, if the coefficient of the variable in the objective function is positive, you increase the variable. Else, if its coefficient is negative you decrease the variable.

Simplex method – Algebraic form

Problem as we transformed it in the previous slide

Maximize

$$z = 6x + 5y$$

Subject to the constraints (notice we have rewritten the constraints as in the previous slide):

$$s = 5 - x - y$$

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and $x, y, s, t \geq 0$

- Recall, currently $x = 0$ and $y = 0$. Therefore the objective function, substituting these values, is $z = 0$. Can we improve z ?
- Yes. We can either **increase*** the value of x or **increase*** the value of y .

*To increase the objective function, if the coefficient of the variable in the objective function is positive, you increase the variable. Else, if its coefficient is negative you decrease the variable.

Simplex method – Algebraic form

Problem as we transformed it in the previous slide

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- Recall, currently $x = 0$ and $y = 0$. Therefore the objective function, substituting these values, is $z = 0$. Can we improve z ?
- Yes. We can either **increase*** the value of x or **increase*** the value of y .
- In simplex, you improve only **one variable** at a time.

*To increase the objective function, if the coefficient of the variable in the objective function is positive, you increase the variable. Else, if its coefficient is negative you decrease the variable.

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:
 - ▶ from $x = 0$ to $x = 5$ according to the first constraint.

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:
 - ▶ from $x = 0$ to $x = 5$ according to the first constraint. Anything more will violate the non-negativity constraint for the slack variable s . Or

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:
 - ▶ from $x = 0$ to $x = 5$ according to the first constraint. Anything more will violate the non-negativity constraint for the slack variable s . Or
 - ▶ from $x = 0$ to $x = 4$ according to the second constraint.

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:
 - ▶ from $x = 0$ to $x = 5$ according to the first constraint. Anything more will violate the non-negativity constraint for the slack variable s . Or
 - ▶ from $x = 0$ to $x = 4$ according to the second constraint. Anything more will violate the non-negativity constraint for t .

Simplex method – Algebraic form

Problem as it stands!

Maximize

$$z = 6x + 5y$$

Subject to the constraints

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

and $x, y, s, t \geq 0$

- Choose to increase x (since it has a larger coefficient of the two choices, hence gives higher increase of the objective).
- However, having chosen to increase x , we can either increase it:
 - ▶ from $x = 0$ to $x = 5$ according to the **first** constraint. Anything more will violate the non-negativity constraint for the slack variable s . Or
 - ▶ from $x = 0$ to $x = 4$ according to the **second** constraint. Anything more will violate the non-negativity constraint for t .
- Choosing the **minimum feasible** of $x = 5$ and $x = 4$ will ensure that both $s, t \geq 0$.

Simplex method – Algebraic form

- Since we have chosen the **minimum increase** of x from $x = 0$ to $x = 4$, x enters the basis, while t exits the basis ($x = 4, \implies t = 0$).

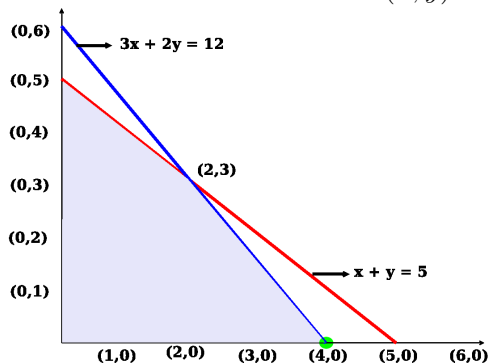
Simplex method – Algebraic form

- Since we have chosen the **minimum increase** of x from $x = 0$ to $x = 4$, x enters the basis, while t exits the basis ($x = 4, \implies t = 0$).
- The new set of **non-basic variables** are therefore y and t , and **basic variables** are x and s . That is, $x = 4, s = 1, y = 0, t = 0$.

Simplex method – Algebraic form

- Since we have chosen the **minimum increase** of x from $x = 0$ to $x = 4$, x enters the basis, while t exits the basis ($x = 4, \implies t = 0$).
- The new set of **non-basic variables** are therefore y and t , and **basic variables** are x and s . That is, $x = 4, s = 1, y = 0, t = 0$.

Visualize what this solution of $(x, y) = (4, 0)$ implies



Simplex method – Algebraic form

- Since we have chosen the **minimum increase** of x from $x = 0$ to $x = 4$, x enters the basis, while t exits the basis ($x = 4, \implies t = 0$).
- The new set of **non-basic variables** are therefore y and t , and **basic variables** are x and s . That is, $x = 4, s = 1, y = 0, t = 0$.

Rewrite the constraints in terms of the new basic variables

Rewrite the second constraint ($t = 12 - 3x - 2y$), now with x as the **basic variable**:

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

The first constraint in the previous iteration was $s = 5 - x - y$. Rewriting this by substituting x from above, becomes:

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

Simplex method – Algebraic form

Problem as it is now transformed – second iteration

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

Objective was previously to maximize:

$$z = 6x + 5y$$

now becomes (in terms of the new **non-basic** variables y and t)

$$z = 24 + y - 2t$$

Simplex method – Algebraic form

Problem as it is now transformed – second iteration

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

Objective was previously to maximize:

$$z = 6x + 5y$$

now becomes (in terms of the new **non-basic** variables y and t)

$$z = 24 + y - 2t$$

Notice how the objective function is growing

Previously, $z = 0$ (with x and y as the non-basic variables). Now, $z = 24$ when $y = 0$ and $t = 0$ are the non-basic variables

Simplex method – Algebraic form

Problem as it is now transformed – second iteration

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

Objective was previously to maximize:

$$z = 6x + 5y$$

now becomes (in terms of the new **non-basic** variables y and t)

$$z = 24 + y - 2t$$

Notice how the objective function is growing

Previously, $z = 0$ (with x and y as the non-basic variables). Now, $z = 24$ when $y = 0$ and $t = 0$ are the non-basic variables

Can z be increased further?

Simplex method – Algebraic form

(Note: What we are doing here is a **repetition** of the steps/logic undertaken in the previous iteration)

- To increase z further, we can either **increase** y (since coefficient in the objective is **positive**) or **decrease** t (since the coefficient in the objective is **negative**).
- Decreasing t is **infeasible** because t is already zero and any further decrease will **violate** non-negativity constraint. Therefore, our only choice here is to **increase** y
- However, y can only be increased from $y = 0$ to $y = 3$ (based on the second equation) for the solution to be feasible.

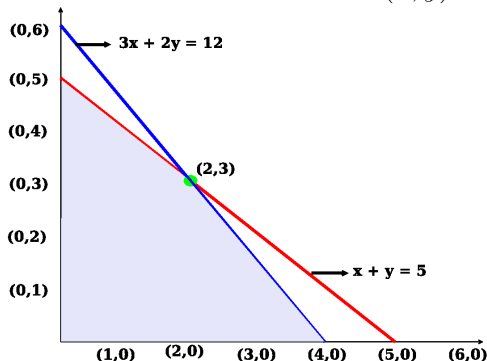
Simplex method – Algebraic form

Since we have chosen to **increase y from 0 to 3**, the new **basic variables** are x, y whose current values are $x = 2$ and $y = 3$ and the **non-basic variables** are $s = 0, t = 0$.

Simplex method – Algebraic form

Since we have chosen to **increase y from 0 to 3**, the new **basic variables** are x, y whose current values are $x = 2$ and $y = 3$ and the **non-basic variables** are $s = 0, t = 0$.

Visualize what this solution of $(x, y) = (2, 3)$ implies:



Simplex method – Algebraic form

Since we have chosen to **increase y from 0 to 3**, the new **basic variables** are x, y whose current values are $x = 2$ and $y = 3$ and the **non-basic variables** are $s = 0, t = 0$.

Rewrite the constraints in terms of the new basic variables

we rewrite the second equation ($s = 1 - \frac{1}{3}y + \frac{1}{3}t$) with y as the **basic variable**:

$$y = 3 - 3s + t$$

But, the first equation in the iteration is $x = 4 - \frac{2}{3}y - \frac{1}{3}t$. Rewriting this by substituting y from above, we get:

$$x = 2 + 2s - t$$

Simplex method – Algebraic form

Problem as it is now transformed

$$x = 2 + 2s - t$$

$$y = 3 - 3s + t$$

Objective was to maximize:

$$z = 24 + y - 2t$$

now becomes (in terms of new **non-basic** variables)

$$z = 27 - 3s - t$$

Simplex method – Algebraic form

Problem as it is now transformed

$$x = 2 + 2s - t$$

$$y = 3 - 3s + t$$

Objective was to maximize:

$$z = 24 + y - 2t$$

now becomes (in terms of new **non-basic** variables)

$$z = 27 - 3s - t$$

Observe that $z = 27$ when $s = 0$ and $t = 0$ as the new **non-basic variables**

Simplex method – Algebraic form

Problem as it is now transformed

$$x = 2 + 2s - t$$

$$y = 3 - 3s + t$$

Objective was to maximize:

$$z = 24 + y - 2t$$

now becomes (in terms of new **non-basic** variables)

$$z = 27 - 3s - t$$

Observe that $z = 27$ when $s = 0$ and $t = 0$ as the new **non-basic variables**
Can z be increased further?

Simplex method – Algebraic form

Problem as it is now transformed

$$x = 2 + 2s - t$$

$$y = 3 - 3s + t$$

Objective was to maximize:

$$z = 24 + y - 2t$$

now becomes (in terms of new **non-basic** variables)

$$z = 27 - 3s - t$$

Observe that $z = 27$ when $s = 0$ and $t = 0$ as the new **non-basic variables**
Can z be increased further?

The only way to increase z further is to either **decrease** s and/or **decrease** t .
Both both these choices are **infeasible** Therefore **STOP!**

Simplex method – Algebraic form

When simplex method stopped, the values of **basic variables** were $x = 2$ and $y = 3$ (we get these by substituting $s = 0$ and $t = 0$ (**non-basic variables**) into the constraints in the previous slide.

These are the **optimum values** of the decision variables x and y for this example problem.

Visualize what this solution of $(x, y) = (2, 3)$ implies:

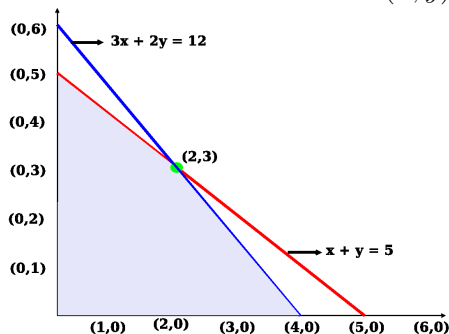


Tableau form of Simplex does **exactly** the same thing but in a convenient tabular way (that can be converted into a computer program)

Now let's see the same simplex method now in **tableau form** (This is now a crank-turning exercise).

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Tableau form

First convert the problem into slack form:

$$z = 6x + 5y$$

$$x + y \leq 5$$

$$3x + 2y \leq 12$$

$$x, y \geq 0$$

becomes,

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Next we list the coefficients C_j of the variables in the objective function:

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$



	C_j	6	5	0	0		
		x	y	s	t		

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Now list the basic variables and their coefficients in the objective function, recalling that we always start with the slack variables in the basis.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$



	Cj	6	5	0	0		
		x	y	s	t		
0	s						
0	t						

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Write down the coefficients on the left-hand-side of the constraints...

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$



	cj	6	5	0	0		
		x	y	s	t		
0	s	1	1	1	0		
0	t	3	2	0	1		

Simplex – Tableau Form (Watch the lecture recording going through these slides)

...and the right-hand-side of the constraints.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

$C_j - Z_j$ are the coefficients of the variables in the objective function when expressed in terms of the **non-basic** variables.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	
	Cj-Zj	6	5	0	0		

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Compute the current value of the objective function.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	
	Cj-Zj	6	5	0	0	Z = 0	



Simplex – Tableau Form (Watch the lecture recording going through these slides)

We always choose to increase the **non-basic** variable with the largest positive coefficient in the objective function.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	
	Cj-Zj	6	5	0	0	Z = 0	



Simplex – Tableau Form (Watch the lecture recording going through these slides)

Compute the maximum value θ that x can be increased to based on each constraint, and choose the minimum feasible of such values in order to satisfy both constraints.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Now x will enter the basis and t will leave the basis...

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	



Simplex – Tableau Form (Watch the lecture recording going through these slides)

...and we need to rewrite our constraint so that the coefficient of x is 1 as it is now in the basis rather than t . Note that this is equivalent to rewriting the constraint as $x = 4 - \frac{2}{3}y - \frac{1}{3}t$ as we did in the algebraic method.

$$z = 6x + 5y + 0s + 0t$$

$$x + y + s = 5$$

$$x + \frac{2}{3}y + \frac{1}{3}t = 4$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-zj	6	5	0	0	Z = 0	
0	s						
→ 6	x	1	2/3	0	1/3	4	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Similarly, we must express the $x + y + s = 5$ in terms of the new **non-basic** variables (y, t) by substituting in our expression for the new basic variable x .

$$z = 6x + 5y + 0s + 0t$$

$$\frac{1}{3}y + s - \frac{1}{3}t = 1$$

$$x + \frac{2}{3}y + \frac{1}{3}t = 4$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-zj	6	5	0	0	Z = 0	
→ 0	s	0	1/3	1	-1/3	1	
6	x	1	2/3	0	1/3	4	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Finally, we need to express the objective function ($z = 6x + 5y + 0s + 0t$) in terms of the new **non-basic** variables (y, t).

$$\begin{aligned}
 z &= 24 + y - 2t \\
 \frac{1}{3}y + s - \frac{1}{3}t &= 1 \\
 x + \frac{2}{3}y + \frac{1}{3}t &= 4 \\
 x, y, s, t &\geq 0
 \end{aligned}$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	
6	x	1	2/3	0	1/3	4	
	Cj-Zj	0	1	0	-2	Z = 24	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

We are again ready to move to the next iteration and so we see if the objective function can be increased any further. The only option we have is to increase y .

Think about why shouldn't we increase t .

$$z = 24 + y - 2t$$

$$\frac{1}{3}y + s - \frac{1}{3}t = 1$$

$$x + \frac{2}{3}y + \frac{1}{3}t = 4$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	
6	x	1	2/3	0	1/3	4	
	Cj-Zj	0	1	0	-2	Z = 24	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Compute the maximum value θ that y can be increased to based on each constraint, and choose the minimum of such values in order to satisfy both constraints.

$$z = 24 + y - 2t$$

$$\frac{1}{3}y + s - \frac{1}{3}t = 1$$

$$x + \frac{2}{3}y + \frac{1}{3}t = 4$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	3
6	x	1	2/3	0	1/3	4	6
	Cj-Zj	0	1	0	-2	Z = 24	



Simplex – Tableau Form (Watch the lecture recording going through these slides)

Now y enters the basis and s leaves the basis and we rewrite the first constraint
 $\frac{1}{3}y + s - \frac{1}{3}t = 1$ so that the coefficient of y is 1.

$$z = 24 + y - 2t$$

$$y + 3s - t = 3$$

$$x + \frac{2}{3}y + \frac{1}{3}t = 4$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	3
6	x	1	2/3	0	1/3	4	6
	Cj-Zj	0	1	0	-2	Z = 24	
→ 5	y	0	1	3	-1	3	

Simplex – Tableau Form (Watch the lecture recording going through these slides)

Similarly, we must express the $x + \frac{2}{3}y + \frac{1}{3}t = 4$ in terms of the new **non-basic** variables (s,t) by substituting in our expression for the new basic variable y .

$$z = 24 + y - 2t$$

$$y + 3s - t = 3$$

$$x - 2s + t = 2$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	0
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	3
6	x	1	2/3	0	1/3	4	6
	Cj-Zj	0	1	0	-2	Z = 24	
5	y	0	1	3	-1	3	
6	x	1	0	-2	1	2	



Simplex – Tableau Form (Watch the lecture recording going through these slides)

Finally, we express the objective function ($z = 6x + 5y$) in terms of (s, t) and we can see that the objective function cannot be increased further without violating our constraints and so we stop.

$$z = 27 - 3s - t$$

$$y + 3s - t = 3$$

$$x - 2s + t = 2$$

$$x, y, s, t \geq 0$$

	Cj	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	5
0	t	3	2	0	1	12	4
	Cj-Zj	6	5	0	0	Z = 0	
0	s	0	1/3	1	-1/3	1	3
6	x	1	2/3	0	1/3	4	6
	Cj-Zj	0	1	0	-2	Z = 24	
5	y	0	1	3	-1	3	
6	x	1	0	-2	1	2	
→	Cj-Zj	0	0	-3	-1	Z = 27	

Converting a non-standard LP into a Canonical Form – not examinable!

In general, linear programming formulations often present themselves in a **non-standard** form, as against the **canonical** formulation handled in this lecture. Often simple algebraic manipulations will change a non-standard LP formulation to a canonical form, such as those given in the table below:

Non-Standard LP form	Example (Non-Standard LP form)	Canonical LP form requirement	Conversion Rule	Example (Canonical)
min objective	$\min z = 4x + 7y$	max objective	Multiply objective by -1	$\max -z = -4x - 7y$
$\geq$ constraint	$3x + 2y \geq 13$	$\leq$ constraint	Multiply constraint by -1	$-3x - 2y \leq -13$
$=$ constraint	$5x + 3y = 14$	$\leq$ constraint	Convert $=$ into two $\leq$ and $\geq$ constraints	$5x + 3y \leq 14$ $-5x - 3y \leq -14$
LP variable $\geq a$	$x \geq 5$	≥ 0 implicit constraint	Variable substitution	$x' = x - 5 \geq 0$
LP variable $\leq b$	$x \leq 5$	≥ 0 implicit constraint	Make it an explicit constraint	$x \leq 5$ (as explicit constraint)
unconstrained LP variable	unconstrained x	≥ 0 implicit constraint	Variable substitution	$x = x^+ - x^-$ $x^+, x^- \geq 0$
LP variable ≤ 0	$x \leq 0$	≥ 0 implicit constraint	Variable substitution	$x' = -x \geq 0$

--o0o--
END
--o0o--